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Microsoft Agent Framework



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Agenda

- Core Concepts & Building Blocks
  - Agents
  - Workflows
- DEMOs (with MCP tool usages, state-management with AgentThread, and workflows...)

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Core Concepts & Building Blocks

Agents

Agents are systems that combine reasoning, context and tools to pursue objectives.

Workflows

Workflows structure complex objectives into sequences of steps, coordinating tasks across people or systems to reach a goal efficiently.

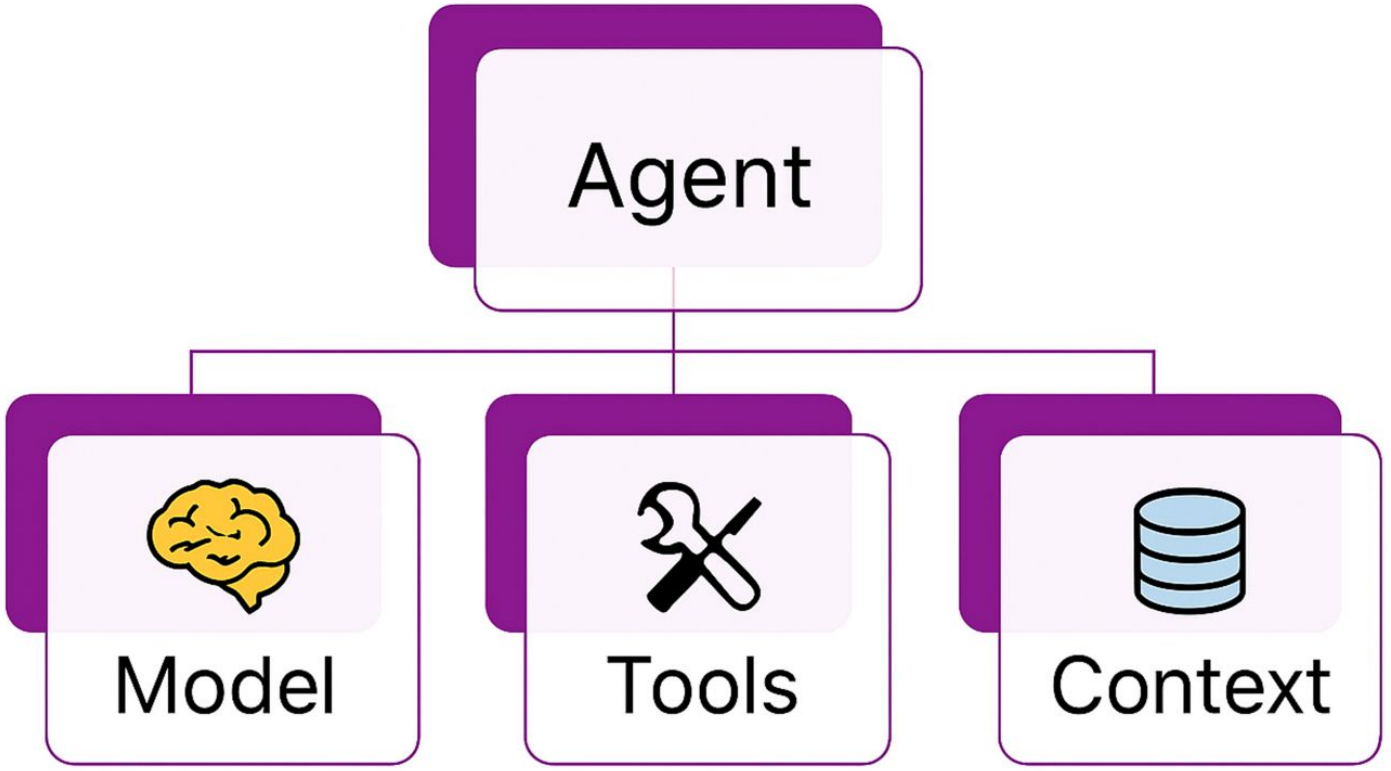
State Management (AgentThread)

AgentThread is the state manager that carries an agent's conversation history so it can reason consistently across turns.

Tools / Plugins

The capabilities. Functions or MCP Servers that allow agents to interact with the real world.

Core Concepts: Agents



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```
var chatClient = new OpenAIClient(
    new ApiKeyCredential(apiKey!),
    new OpenAIClientOptions
    {
        Endpoint = new Uri(baseUrl),
    }
).GetChatClient("gpt-5.2");

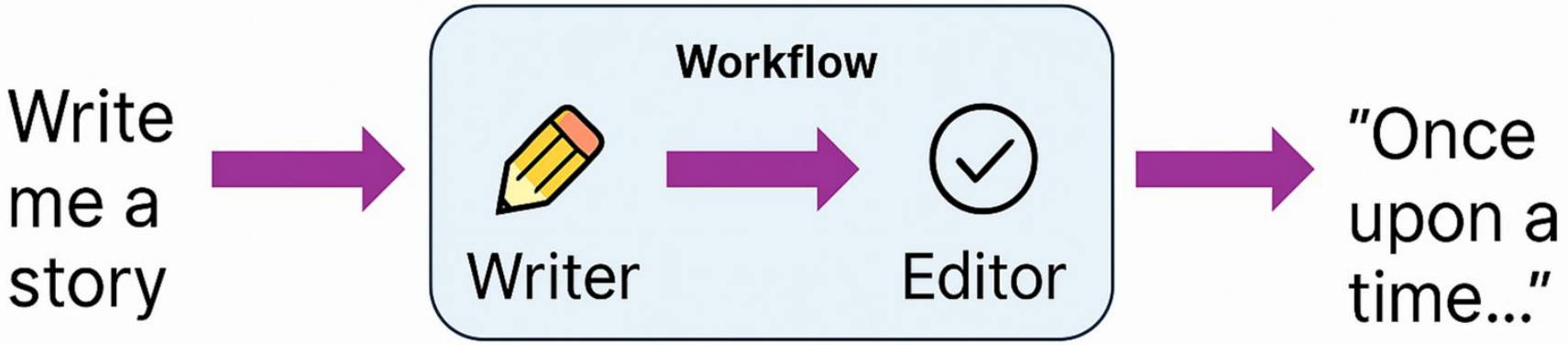

AIAgent agent = chatClient.CreateAIAgent(
                    instructions: @"You are a helpful GitHub assistant.
                                   You can answer questions about GitHub repositories,
                                   commits, issues, and pull requests.
                                   Always provide clear and concise information.",
                    tools: [.. mcpTools.Cast<AITool>()]
                );
```

Individual agents that use LLMs to process user inputs, make decisions, call tools and MCP servers to perform actions, and generate responses.

- **Agents** are wrappers around ChatClient.
- They support **streaming** and **structured** outputs.

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Core Concepts: Workflows



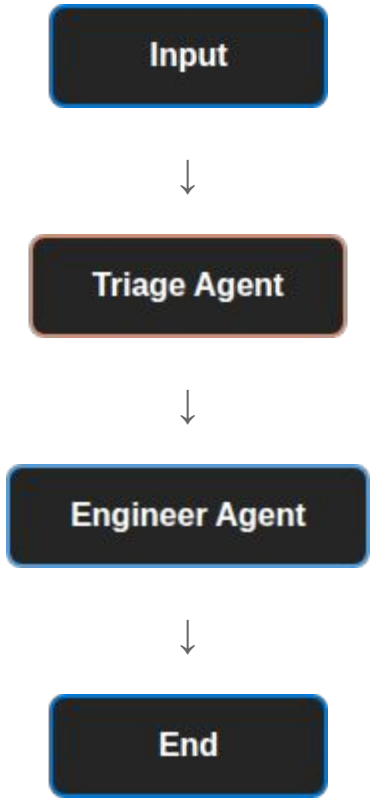
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```
WorkflowBuilder b = new();

// Define Flow: Input -> Triage -> Engineer
b.AddEdge(inputNode, triageAgent);
b.AddEdge(triageAgent, engineerAgent);

var workflow = b.Build();

await workflow.RunAsync("Fix the bug");
```



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DEMO: MCP Usage & State Management with AgentThread

1-) MCP Integration: Connects to Github's MCP Server using **@modelcontextprotocol/server-github** NPM package. The agent automatically discovers all available Github tools (commits, issues, PRs) and uses them dynamically (no manuel tool definition needed).

2-) AgentThread for State Management: Maintains conversation context across multiple queries. It answers directly from memory without making another Github API Call, or any other tool calls!

Source Code: https://github.com/EngincanV/samples/tree/main/IssueCopilotDemo

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DEMO: Release Notes Workflow

1-) Writer Agent: Technical Release Notes Writer

2-) Reviewer Agent (Marketing Agent): Polishes the technical release notes and makes it customer-friendly!

Source Code: https://github.com/EngincanV/samples/tree/main/ReleaseNotesWorkflowDemo

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Thanks for listening!